

Chapter 3: Dynamics 1: Mass and weight

Force (definition)

- A force is a push or pull that one object exerts on another.

- Force can cause an object to:

→ start or stop moving

→ change the speed of object

→ change direction

→ change shape of object

- SI unit → Newton (N)

- Vector quantity

Types of Force

Contact Force

- Normal contact force

- Frictional Force

- Tension Force

- Air resistance

→ exists between objects

that are in contact

Non-contact force

- Gravitational force (aka weight)

- Electrostatic force

- Magnetic force

→ do not require objects to be in contact to exist

Contact Forces

1. Normal contact force

- The force exerted by a surface on an object pressing on it.

- The force is always perpendicular to the surface.

2. Frictional force

- The force that opposes motion between two surfaces in contact.

- The force is parallel to the surface, opposite to the direction of motion.

3. Tension Force

- The pull exerted by a stretched string/rope on an object attached to it.
- Direction of tension force is always acting away from the object.

4. Air Resistance

- The frictional force exerted by air that opposes the motion of moving objects.
- The direction of air resistance is opposite direction of motion.

Non-Contact Forces

1. Gravitational Force / Weight

- The pull exerted by Earth's gravity on any object.

2. Electrostatic Force

- The force between electric charges.
- Attractive electric charges between unlike charges.
- Repulsive electric charges between like charges.

3. Magnetic Force

- The force between magnets.
- Attractive magnetic forces between unlike poles.
- Repulsive ~~etc~~ magnetic forces between like poles.

Mass and Weight (definition)

- Mass is the measure of the amount of matter in a body.
- Scalar quantity
- SI unit $\rightarrow$ kilogram (kg)
- Weight is the gravitational force acting on an object that has mass
- Vector quantity
- SI unit $\rightarrow$ Newton (N)

Gravitational Field (definition)

- Gravitational field is a region in which a body of mass experiences a force due to gravitational attraction.

Gravitational Field Strength (definition)

- Gravitational field strength (g) is defined as the gravitational force ^{acting} per unit mass on an object.
- Gravitational field strength is measured in N/kg .
- Earth has a gravitational field strength of 10 N/kg .

Relationship between mass and weight

$$\text{Weight (N)} = \text{Mass (kg)} \times \text{Gravitational field strength (N/kg)}$$

$$W = mg$$

Summary: Mass vs weight

Mass	Weight
- Amount of matter in the body	- Force of gravity on the body
- Constant at any location	- Depends on gravitational field strength of the location
- Scalar quantity	- Vector quantity
- Kilograms (kg)	- Newtons (N)
- Measured using beam balance/electronic balance	- Measured using spring balance

Factors affecting gravitational field strength

1. **MASS:** The more massive an object is, the stronger its gravitational field strength.
2. **Distance from object:** The greater the distance from the object, the weaker the gravitational field strength of the object exerted on another body.

Chapter 4: Dynamics 2: Forces

Balanced and unbalanced forces

Balanced Forces

- When forces are balanced, resultant force = 0N
- The object is in equilibrium. (no resultant force)

Unbalanced Forces

- When forces are unbalanced, resultant force $\neq$ 0N
- The object moves in the direction of the larger force.

Newton's First law of motion (definition)

Newton's First Law of Motion states that an object will remain at rest or continue to move in a straight line at constant speed, unless a resultant force acts on it.

- When an object at rest is subjected to balanced forces, object will remain at rest.
- When a moving object is subjected to balanced forces, it will continue to move at constant speed in a straight line. (constant velocity, zero acceleration)

Keywords to tell if it is 1st Law:

- "remains stationary" $\Rightarrow$ no resultant force $\Rightarrow$ balanced forces
- "constant speed" / "zero acceleration" $\Rightarrow$ no resultant force $\rightarrow$

Inertia (definition)

The inertia of an object refers to the reluctance of the object to change its state of rest or motion due to its mass.

- The larger the mass, the larger the inertia.
 - $\hookrightarrow$ i.e. The larger the mass, the harder it is for the object to start moving, slow down, move faster or change direction.

Newton's second law of motion (definition)

Newton's second law of motion states that when a resultant force acts on an object of constant mass, the object will accelerate in the direction of the resultant force.

- The resultant force acting on an object is given by:

$$F_{\text{net}} = ma$$

(N) (kg) (m/s^2)

- Since m is a constant, F_{net} is directly proportional to a .

Resultant Force

- A resultant force will cause:

a. A stationary object to move

b. A moving object to:

- stop moving
- move faster (acceleration)
- move slower (deceleration)
- change direction

When to use 2nd law:

- When 2 of the 3 of F_{net} , m or a is given and you need to find the unknown.

Newton's third law of motion (definition)

Newton's Third law of motion states that for every action force, there is an equal and opposite reaction force, and these forces act on mutually opposite bodies.

(these forces must be the same type of force)

Example:

When a person stands on the ground, the person exerts a normal contact force on the ground in downward direction. The ground exerts a normal contact force on the person of equal magnitude but upward direction.

Free Body Diagrams

- A Free body diagram is a schematic diagram that shows all the forces acting on the object of interest.

Step 1: Identify object of interest.

Step 2: Do a quick sketch of the object.

Step 3: Identify possible forces that can act on the object.

Common forces: Normal contact force, weight, frictional force, tension force, resistive force (air resistance)

Take note when labelling forces:

1. Weight must start from center of object.
2. Normal contact force / pushing forces start/end at the surface of contact, not in between two surfaces.
3. Frictional force must be drawn on the surface of contact.
4. Tension force must be drawn on the string/rope etc., away from object.

Vector Diagrams

- To add more than 2 vectors, use the tip-to-tail method.
- The resultant vector is the tail of the first vector to the tip of the last vector.
- If the forces form a closed loop (i.e. no resultant force), the object is in equilibrium (balanced forces).

Friction (definition)

- Friction is the contact force that opposes or tends to oppose motion between surfaces in contact.
- Friction is the result of irregularities between surfaces.
- It depends on the material and roughness of surfaces.
- It is not dependent on area of contact.
- Also depends on weight ($\uparrow$ weight = $\uparrow$ frictional force)

Effects of friction

Positive Effects

1. We can walk without slipping.
2. Moving objects are able to slow down when necessary.

Negative Effects

1. Causes wear and tear.
2. Slower motion as friction opposes motion.

Reducing Negative Effects of Friction

- Using wheels
- Using ball bearings
- Using lubricants & polished surfaces
- Using air cushion

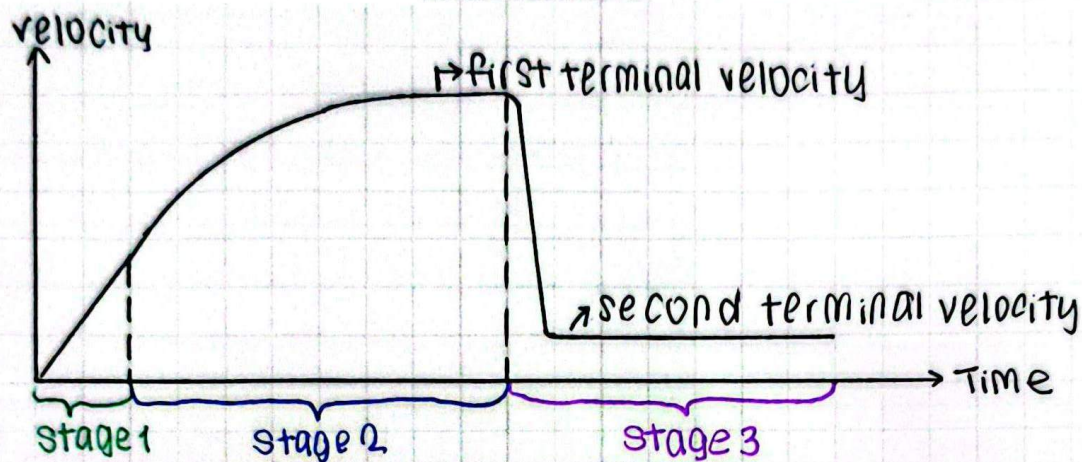
Enhancing Positive Effects of Friction

- Using treads on tyres
- Using parachutes when skydiving

Air Resistance

- When an object is moving downwards through air, the object experiences a force opposing its motion due to the air.
- As the velocity of the object increases, the air resistance increases. (vice-versa)
- Larger surface area ^{of object} increases air resistance.

Motion of sky diver with parachute (downward motion)



Stage 1

- Skydiver jumps out of the aircraft.
- Velocity of skydiver is small = air resistance negligible (free fall)
- Skydiver moves down with constant acceleration of 10 m/s^2 .
- Velocity of skydiver increases.

Using Newton's Second Law,

$$F_{\text{net}} = ma$$

$$W = ma \rightarrow \text{Should be } W - F_{\text{air}} = ma, \text{ since } F_{\text{air}} = 0, \text{ so } W = ma$$

$$mg = ma$$

$$\therefore g = a \rightarrow \text{acceleration} = \text{gravitational field strength} = 10 \text{ m/s}^2$$

Stage 2

- As velocity increases, air resistance increases.
- Acceleration decreases as air resistance increases.
(see below)

Using Newton's Second Law,

$$F_{\text{net}} = ma$$

$$W - F_{\text{air}} = ma \rightarrow \text{When } F_{\text{air}} \text{ increases, } W - F_{\text{air}} \text{ decreases.}$$

So, a decreases. (since m is a constant)

- Air resistance continues to increase until $W = F_{\text{air}}$,
and acceleration = 0 m/s^2 . (first terminal velocity
→ constant velocity)

Stage 3

- Parachute opens.
- There is a large increase in air resistance.
- Since air resistance $>$ weight, the skydiver undergoes deceleration, and velocity decreases.

Using Newton's Second Law,

$$F_{\text{net}} = ma$$

$$W - F_{\text{air}} = ma$$

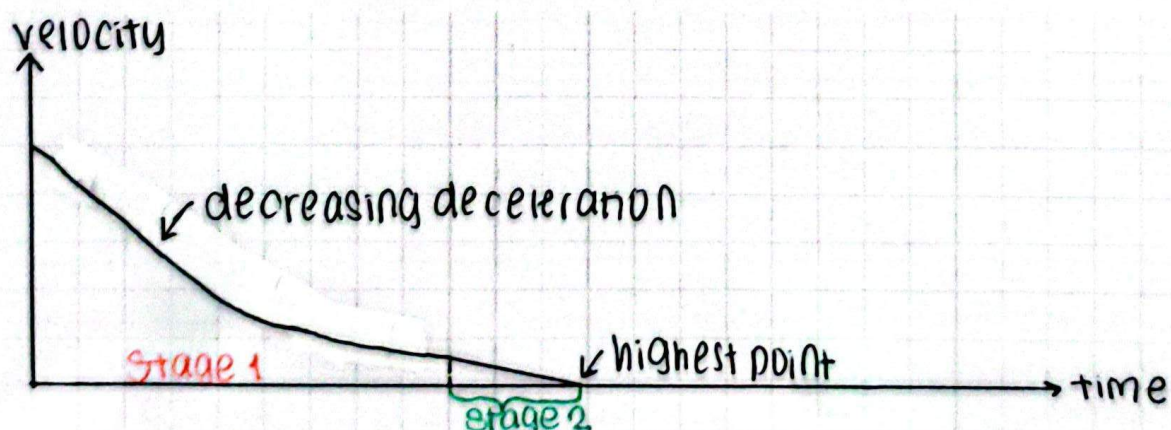
Since $F_{\text{air}} > W$ i.e. $W - F_{\text{air}} < 0$, $a < 0$ (aka deceleration)

- As velocity decreases, air resistance decreases, until $W = F_{\text{air}}$, and acceleration $= 0 \text{ m/s}^2$. (second terminal velocity)
- The skydiver lands with this lower constant velocity.

Terminal velocity (definition)

- Terminal velocity occurs when the object starts to travel at a constant speed where the air resistance against the object is equal to its weight.
- Object has zero acceleration.

Motion of object thrown upwards (upward motion)



Stage 1

- Object experiences both weight and air resistance in downward direction
 - ↳ deceleration $> 10 \text{ m/s}^2$
- At the start of upward motion, velocity of object is the largest. Air resistance is also largest. Deceleration is at its largest value.
- Velocity of the object decreases. The air resistance experienced by the object decreases. Deceleration decreases.

Stage 2

- Velocity of the object decreases until air resistance is negligible.
- The velocity of the object is 0 m/s and ^{object is} at its highest point.
- The acceleration of the object is 10 m/s^2 downward. (Free fall)

No air resistance

- When there is no air resistance and the same experiment is repeated, the object will take longer to reach highest point and will reach a higher point.
- Since there is no air resistance, the object decelerates at a constant rate, deceleration $= 10 \text{ m/s}^2$.
- Therefore, deceleration without air resistance $<$ deceleration with air resistance